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SEPTEMBER 10, 1956
SUPERSEDING
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October 22, 1953

FEDERAL SPECIFICATION

LEAK PREVENTIVE COMPOUND, RADIATOR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of leak preventive compound for use in the cooling system of internal combustion engines (see 6.1).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATION

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

O–E–771—Ethylene Glycol, Inhibited.
RR–S–366—Sieves, Standard for Testing Purposes.
PPP–C–96—Cans, Metal, 28 Gage and Lighter.

Federal Test Method Standard:

No. 791—Lubricants, Liquid Fuels, and Related Products, Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication.—The following Uniform Freight Classification publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

Uniform Freight Classification Rules.

(Application for copies of Uniform Freight Classification Rules should be addressed to the Uniform Freight Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

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FSC Class 6850

3. REQUIREMENTS

3.1 Material.—The cooling system leak preventive compound shall consist of a volatile solution containing a filler, a water-dispersible binder, and additives, such as preservatives and rust inhibitors, required to prevent deterioration of the compound within the container and to protect the cooling system metals. The material shall be nonabrasive and shall be free of silicates, or materials subject to organic or bacterial attack for which no preservative is provided, and free of any other material which may have a harmful effect on the radiator or other components of the cooling system of an engine. The compound shall not clog the tubes of the radiator, and shall be readily removable from the cooling system by means of cooling system cleaners.

3.2 Physical and chemical requirements.—The compound shall conform to the physical and chemical requirements specified hereafter.

3.2.1 *Volatile matter.*—The evaporation loss of the compound shall not exceed 90 percent, by weight, when determined in accordance with 4.3.1.

3.2.2 *Solubility of nonvolatile matter.*—When mixed with water and allowed to stand for 2 hours, the nonvolatile matter of the compound shall not form into lumps, nor leave any hard residue in the test container, when tested in accordance with 4.3.2.

3.2.3 *pH value of mixtures of the compound.*

3.2.3.1 *Water mixture.*—The pH value of a mixture of the compound and distilled water shall be in the range of 6.0 to 9.5, when tested in accordance with 4.3.3.1.

3.2.3.2 *Inhibited ethylene glycol mixture.*—The pH value of a mixture of the compound and borax-inhibited ethylene glycol solution shall be in the range of 7.0 to 8.0, when tested in accordance with 4.3.3.2.

3.2.4 *Dispersibility in water and inhibited ethylene glycol.*—The compound shall be dispersible in water and in borax-inhibited ethylene glycol solutions. The mixtures of the compound with these substances shall show no gelling or gumming, and all phases of the mixtures shall pass through a U. S. No. 16 standard test sieve (not more than 20 separate small particles of the filler may be retained on the sieve), when tested in accordance with 4.3.4.1 and 4.3.4.2.

3.2.5 *Leak stoppage.*—The compound shall effectively stop leakage through $\frac{1}{32}$-inch diameter holes, and shall hold under a pressure of 15 p.s.i., when the compound is employed with water and inhibited glycol-water solutions when tested in accordance with 4.3.5.

4. SAMPLING, INSPECTION, AND TEST PROCEDURE

4.1 Sampling.

4.1.1 *Lot.*—For purposes of sampling, a lot shall consist of all the compound produced by one manufacturer, and offered for delivery at one time.

4.1.2 *Sampling for inspection of filled containers.*—A random sample of filled containers shall be taken from each lot by the inspector in accordance with Military Standard MIL–STD–105 at Inspection Level I and Acceptable Quality Level (AQL) = 2.5 percent defective, to verify compliance with all stipulations of this specification regarding fill, closure, marking, and other requirements not involving tests.

4.1.3 *Sampling for tests.*—A minimum of four containers shall be taken from the lot for test purposes.

4.2 Inspection.—Unless otherwise specified, inspection shall be at the point of manufacture. All inspection and testing shall be made under the supervision of the inspector.

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4.3 **Tests.** — The test methods specified hereafter shall be used in determining compliance of the material under test with the corresponding requirement specified in section 3 of this specification.

4.3.1 *Volatile matter.*—Place approximately 20 grams of well-mixed compound, weighed to the nearest 0.1 gram, in a tared petri dish (approximately 100 mm. in diameter and 15 mm. in depth). Place the dish and its contents for 48 hours in a draft-free oven maintained at 210° ± 3°F. At the end of this period remove the dish from the oven, place it in a desiccator, cool to room temperature, and weigh it to the nearest 0.1 gram. Retain the nonvolatile residue for the determination of the solubility (see 4.3.2). Calculate the percentage of volatile matter as follows:

$$\text{Percent volatile matter} = 100\,\frac{(W - R)}{W}$$

in which:

W = initial weight of sample, in grams
R = weight of residue, in grams

4.3.2 *Solubility of the nonvolatile matter.*—Add 25 ml. of distilled water to the nonvolatile residue retained from the volatile matter test (see 4.3.1), and mix thoroughly. Allow the mixture to stand for 2 hours at room temperature. At the end of this period, examine the mixture for formation of lumps and for hard deposits in the test containers.

4.3.3 *pH value of compound mixtures.*

4.3.3.1 *Water mixture.*—Place 5.0 ml. of the well-mixed compound in an 8-ounce screw-cap glass jar. Add 200 ml. of distilled water to the jar; cover it, shake its contents, allow the jar to stand for 10 minutes and filter. Determine the pH value of the filtrate, using the glass electrode and reference calomel electrode specified in method 5106 of Federal Standard No. 791.

4.3.3.2 *Inhibited ethylene glycol mixture.*—Place 5.0 ml. of the well-mixed compound in an 8-ounce screw-cap glass jar. Add 200 ml. of a 1:1 mixture of inhibited ethylene glycol, conforming to type 1 of Federal Specification O–E–771, and distilled water, to the jar; cover it, shake its contents, allow to stand for 10 minutes and filter. Determine the pH value of the filtrate using the glass electrode and reference calomel electrode specified in method 5106 of Federal Standard No. 791.

4.3.4 *Dispersibility.*

4.3.4.1 *In water.*—Place 5.0 ml. of the well-mixed compound in an 8-ounce screw-cap glass jar. Add 200 ml. of distilled water to the jar, cover it, shake its contents, and allow the jar to stand for 10 minutes. Again shake the contents of the jar until thoroughly mixed, and pour it through a U.S. No. 16 standard sieve conforming to Federal Specification RR–S–366. Rinse the jar and sieve with 200 to 300 ml. of distilled water. Observe the sieve for any retained material and for evidence of gumming or gelling of the water-compound mixture.

4.3.4.2 *In inhibited ethylene glycol mixture.*—Place 5.0 ml. of the well-mixed compound in an 8-ounce screw-cap glass jar. Add 200 ml. of a 1:1 mixture of inhibited ethylene glycol, conforming to type I of Federal Specification O–E–771, and distilled water, to the jar, cover it, shake its contents, and allow to stand for 10 minutes. Again shake the contents of the jar until thoroughly mixed, and pour it through a U.S. No. 16 standard sieve conforming to Federal Specification RR–S–366. Rinse the jar and sieve with 200 to 300 ml. of distilled water. Observe the sieve for any retained material and for evidence of gumming or gelling of the ethylene glycol-compound mixture.

4.3.5 *Leak test.*

4.3.5.1 *Apparatus.*—The leak test unit shall consist of a pressurized reservoir, a circulating pump, and an electric heating unit.

4.3.5.1.1 *Reservoir.*—The reservoir shall be approximately 7 by 7 by 8 inches high. Total capacity of the assembled unit should be between 1.7 and 2.0 gallons. The reservoir should be constructed of stainless steel or brass, minimum thickness 0.062 inches, in

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order to withstand a pressure of 20 p.s.i. The top of the reservoir is so constructed that it may be removed for thoroughly cleaning the interior of the reservoir. Two openings having approximately 2½ by 2½ inches of free working area shall be cut in the reservoir. One of the openings shall be centered on the bottom of the reservoir and one centered on the side of the reservoir. Two test sections constructed of brass, not to exceed 0.015 inches in thickness, shall be soldered or bolted over the openings in the reservoir. The bottom test section shall contain one $\frac{1}{32}$-inch diameter hole in the center of the section. The side test section shall contain two $\frac{1}{32}$-inch diameter holes spaced 1½ inches apart on the horizontal center line of the test section. The reservoir should contain a drain on the side or bottom of the reservoir for aid in draining or withdrawing test liquid. The top of the reservoir should have an opening for filling, thermometer well, pressure gauge (0-30 lbs. minimum) and a means of applying air pressure, such as a tire valve stem. The outlet opening or suction connection to the circulating pump is situated near the bottom on a side of the reservoir. The inlet opening or discharge connection of the circulating pump is situated near the top on a side of the reservoir. For ease of construction the inlet and outlet openings should be on adjacent sides of the reservoir. The inlet opening shall be baffled to insure that the liquid discharge is not directed on any of the test openings (see figure 1).

4.3.5.1.2 *Circulation pump.*—The circulation pump shall be capable of circulating a minimum of eight gallons of water per minute against zero head. The packing gland of the pump should be capable of withstanding 20 p.s.i. without excessive leakage. Inlet and outlet connections shall be not less than ¼ inch standard water pipe. The electric motor for the circulation pump should have a power rating of not less than ⅕ hp.

4.3.5.1.3 *Heating element.* — The heating element should be of the immersion cartridge type and should have a power rating between 300 and 500 watts. The heating element functions best when installed in the discharge pipe of the circulation pump. A rheostat or variable transformer should be employed in series with the heating element for controlling water temperatures.

4.3.5.1.4 *Calculation of compound.* — The capacity in gallons of the leak test unit shall be determined by filling with tap water at room temperature. On the basis that the manufacturer's sample is satisfactory for treating a four gallon capacity cooling system, unless otherwise stated, compute the required amount of leak preventive compound for the leak test unit as follows:

$$S = \frac{C}{4} \times U \times 29.6$$

in which:

C = the volume in ounces of the liquid leak preventive compound sample supplied by the manufacturer.

U = the capacity of the test unit, in gallons.

S = the required milliliters of compound to be added to the test unit.

4.3.5.2 *Procedure with water as a coolant.* —The test unit is disassembled and thoroughly cleaned of any remaining leak preventive compound and scale from previous tests. Cleaning of the unit is accomplished by means of scouring powder and a stiff bristle brush. All test openings shall be inspected to insure they are open. The unit is reassembled and filled to overflowing with tap water. All test openings shall show leakage. Withdraw 800 to 1000 milliliters of water. Place the required amount of well-mixed compound under test, determined in accordance with 4.3.5.1.4, in a beaker with 50 ml. of water, and thoroughly mix. Start the circulation pump, and with all test openings leaking, slowly add the leak preventive. All openings shall stop leaking within 120 seconds after all the leak preventive mixture has been added to the reservoir. Circulation is continued for 10 minutes. In the event that overflow is observed due to excessive foam-

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ing, withdraw an additional 500 to 800 ml. of liquid and retain. The side of the reservoir is then struck 3 sharp blows below the test openings and 3 sharp blows above the test openings with a wooden rod, ⅜ to ½ inch in diameter and 16 inches in length. Care shall be taken to refrain from striking closer than 1 inch to the test openings. All leaks shall stop within 120 seconds, slight seepage being permitted. With the reservoir vented to the atmosphere and the circulating pump running, heat is applied until the temperature of the water in the test unit reaches 180° ± 3°F. The side of the reservoir is then struck 3 sharp blows below the test openings and 3 sharp blows above the test openings with the wooden rod. All leaks shall stop leaking within 120 seconds. Circulation is continued for 30 mintes, after which the test unit is shut down and cooled to room temperature within a 16-hour period. At the termination of the 16-hour period, the water level in the reservoir is raised to within 1 inch of the top of the reservoir. (Any retained solution containing leak preventive compound is added at this time). The unit is pressure sealed. The circulation pump is started and heat is applied until the temperature of the water in the reservoir reaches 180° ± 3°F. The pressure in the reservoir is maintained at between 8 and 10 p.s.i. Should any leaks occur during this period, they shall stop leaking within 120 seconds, slight seepage being permitted. When the temperature in the reservoir reaches 180° ±3°F. a pressure of 10 pounds + 0,–1 pound, is maintained for a period of 4 hours. At the termination of the 4 hour period, a pressure of 15 p.s.i. is applied to the reservoir for 1 minute. No leakage shall occur at the test openings. The pressure is lowered to 10 p.s.i. after the 1-minute period, and maintained for 10 minutes. No leakage shall occur at the test openings when the side of the reservoir is struck 3 sharp blows below the test openings and 3 sharp blows above the test openings with the wooden rod.

4.3.5.3 *Procedure with antifreeze as a coolant.*—The test unit is disassembled and thoroughly cleaned of any remaining leak preventive compound and scale from previous tests. Cleaning of the unit is accomplished by means of scouring powder and a stiff bristle brush. All test openings shall be inspected to insure they are open. The unit is reassembled and filled to overflowing with a mixture of one part of packaged inhibited ethylene glycol, conforming to type I of Federal Specification O–E–771, and two parts of tap water. All test openings shall show leakage. Withdraw 800 to 1000 milliliters of the antifreeze coolant. Place the required amount of well-mixed compound under test, determined in accordance with 4.3.5.1.4, in a beaker with 50 ml. of the antifreeze coolant, and thoroughly mix. Start the circulation pump and with all test openings leaking, slowly add the leak preventive mixture. All openings shall stop leaking within 150 seconds after all the leak preventive mixture has been added to the reservoir. Circulation is continued for 10 minutes. In the event that overflow is observed due to excessive foaming withdraw an additional 500 to 800 ml. of the liquid and retain. The side of the reservoir is then struck 3 sharp blows below the test openings and 3 sharp blows above test openings with a wooden rod, ⅜ to ½ inch in diameter and 16 inches in length. Care should be taken to refrain from striking closer than one inch to the test openings. All flow shall stop within 150 seconds, slight seepage being permitted. With the reservoir vented to the atmosphere and the circulation pump running, heat is applied until the temperature of the coolant in the test unit reaches 180° ± 3°F. The side of the reservoir is then struck 3 sharp blows below the test openings and 3 sharp blows above the test openings with the wooden rod. All leaks shall stop leaking within 150 seconds. Circulation is continued for 30 minutes, after which the test unit is shut down and cooled to room temperature within a 16-hour period. The water level in the reservoir is raised to within 1 inch of the top of the reservoir. (Any retained solution containing leak preventive compound is added at this time). The unit is pressure sealed, the circulation pump is

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started, and heat is applied until the temperature of the water in the reservoir reaches 180° ± 3°F. The pressure in the reservoir is maintained at between 8 to 10 p.s.i. Should any leaks occur during this period, they shall stop leaking within 150 seconds, slight seepage being permitted. When the temperature in the reservoir reaches 180° ± 3°F., a pressure of 10 pounds + 0, –1 pound is maintained for a period of 4 hours. At the termination of the 4 hour period, a pressure of 15 p.s.i. is applied to the reservoir for 1 minute. No leakage shall occur at the test openings. The pressure is lowered to 10 p.s.i. after the 1-minute period and maintained for 10 minutes. No flow shall occur at the test openings when the side of the reservoir is struck 3 sharp blows below the test openings and 3 sharp blows above the test openings with the wooden rod.

4.3.6 *Additional tests.*—The right is reserved by the procuring agency to make such additional tests and analyses as it may deem necessary to determine compliance of the material with the requirements of this specification.

5. PREPARATION FOR DELIVERY

(For definitions and levels see 6.3.)

5.1 Packaging.

5.1.1 *Level A.* — The compound shall be packaged in 12-ounce screw-cap cans, conforming to the requirements of Federal Specification PPP–C–96, type VIII. Coating of cans shall be plan A.

5.1.2 *Level C.* — The compound shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 *Level A.*—The compound packaged as specified in 5.1.1, shall be packed in accordance with the overseas shipment requirements of the appendix to Federal Specification PPP–C–96, except that the quantity of cans shall be 24 each. Case liners shall be provided.

5.2.2 *Level B.*—The compound, packaged as specified in 5.1.1, shall be packed in accordance with the domestic shipment requirements of the appendix to Federal Specification PPP–C–96, except that the quantity of cans shall be 24 each. Case liners shall be provided.

5.2.3 *Level C.*—The compound, packaged as specified in 5.1.2, shall be packed in containers of the type, size, and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery to destination. Shipping containers shall comply with the Uniform Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking.—In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use.—The cooling system leak preventive compound is intended as a temporary seal of small pinhole leaks in the radiator and cooling system of internal combustion engines. The compound is not intended to stop seepage of the coolant at the circulation pump shaft.

6.2 Ordering data. — Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of packaging and packing required (see sec. 5).

6.3 Federal Standard No. 102 should be referred to for definitions and applications of

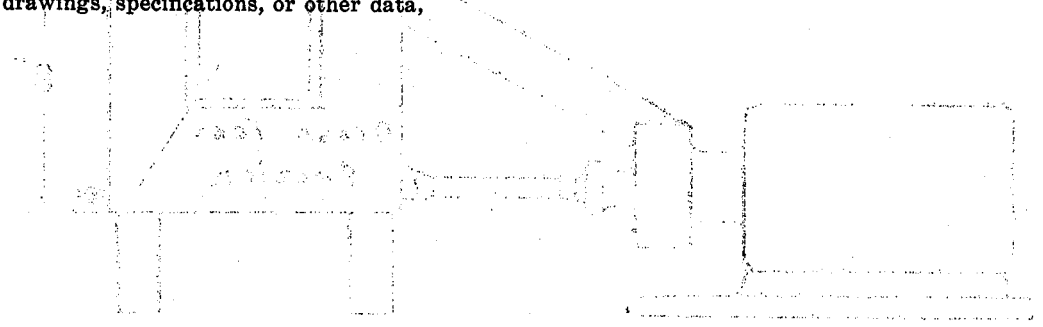
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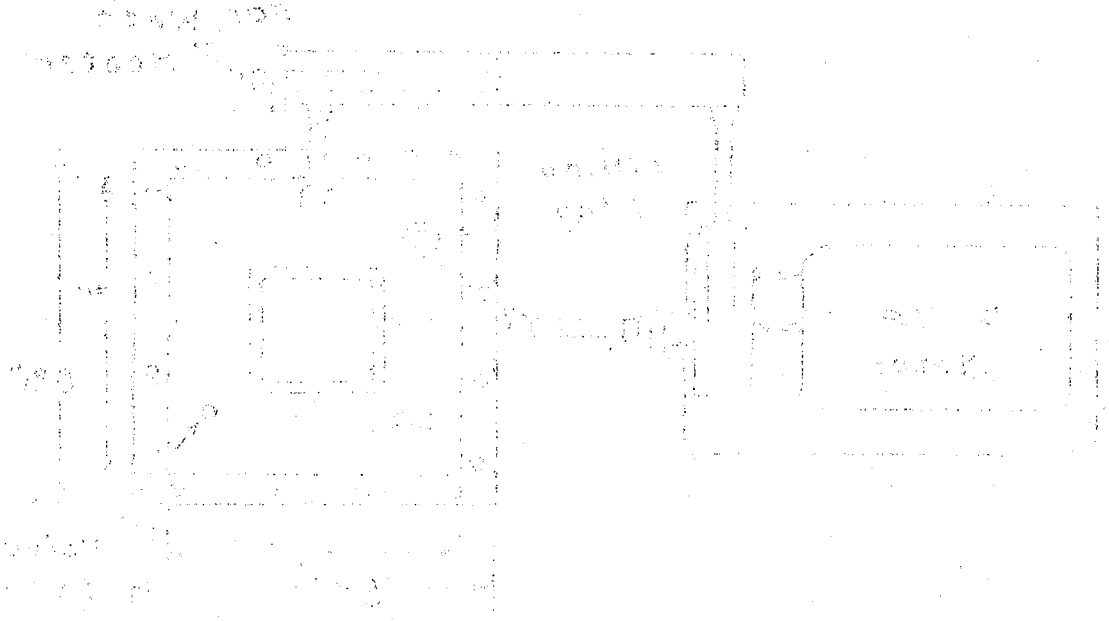
the various levels of packaging protection for supplies and equipment.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:

Army—O E T
Navy—Y MC
Air Force.





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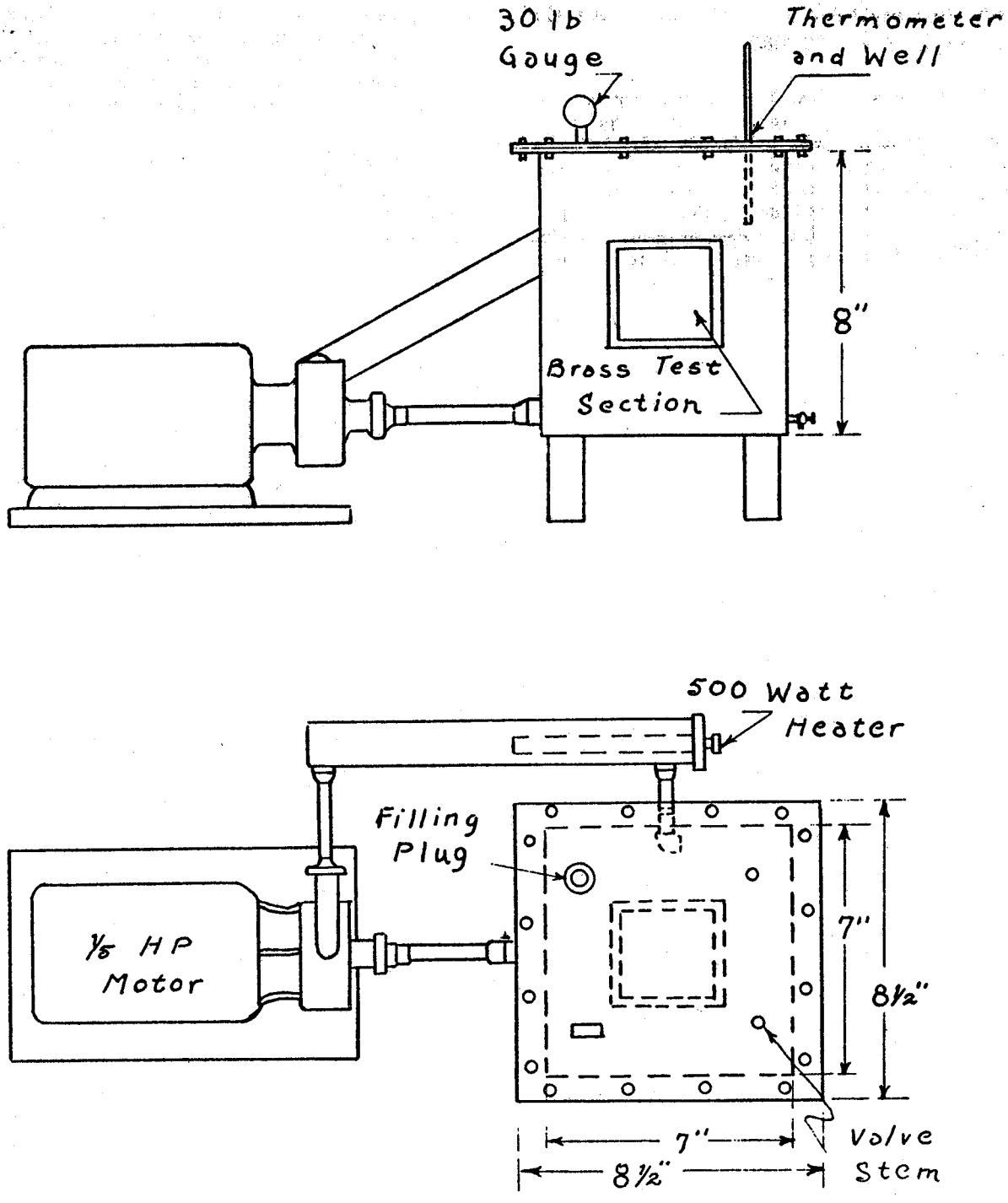


FIGURE 1.

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